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Energy dissipation and behavioral regimes in an autonomous double pendulum subjected to viscous and dry friction damping

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Abstract

The nonlinear dynamics of a system of a compound double pendulum are experimentally analyzed and numerically verified. The experiments are carried out for a combination of initial angles in the range $0 \leq \theta_0, \phi_0 \leq 180^\circ$, with zero initial angular velocity and with an increment of 1° . The motion of the pendula is captured in the video and the individual pendula are tracked using image processing. The energy dissipation in the system is evaluated using the Rayleigh dissipation function to include non-conservative forces as well, as a lumped parameter in the Lagrangian equation. Furthermore, the experimental and the numerical results are brought to agreement by tuning the dissipation factors. The linear, nonlinear and chaotic behavior of the system is classified in the coordinate plane. The behavioral regime represents the dependency of the nature of the system on the total energy and the difference in instantaneous amplitudes of the pendula. Uncertainty in the behavior of the system is observed in the transition region from the nonlinear to the chaotic regime due to the sensitivity of the system to the initial conditions. The dependence of the dissipation factors on the angular displacements is represented as a surface plot, which highlights the dominance of viscosity in the higher amplitude oscillations and the dominance of dry friction in the lower amplitude oscillations. This phenomenon is established by observing the nature of decay in the response plots of the pendula.

Keywords: double pendulum, Rayleigh dissipation function, dissipation factor, viscous damping, dry friction damping

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(Some figures may appear in colour only in the online journal)

1. Introduction

The sensitivity of a system governed by deterministic laws to minuscule changes in the initial conditions leads to its unpredictable and uncertain behavior. This behavior is generally termed as *chaos*, which can be seen in phase plane and phase space analysis. Prediction of the behavior of a system under chaos is extremely difficult, and at the same time, experimental measurements of the responses of such systems are equally laborious and cumbersome, sometimes resulting in erroneous data. Additionally, difficulty is encountered because of the unattainable level of precision in the measurement of the initial conditions [1]. The simplest of the mechanical systems that have been reported to exhibit chaotic behavior is a double-pendulum system. The analysis of the motion of a planar compound double pendulum for different initial conditions opens up a paradigm to understand the linear, the nonlinear and the chaotic dynamics, and hence by extension the behavioral regimes.

Closed-form solutions are available for the single undamped pendulum [2] and the damped pendulum [3], and have also been found for the periodic and the quasiperiodic motion of the double pendulum [4]. Several numerical studies have been documented on the behavior of single-pendulum [5], double-pendulum [6,7] and triple-pendulum systems [8]. A double pendulum exhibits chaotic behavior in a four-dimensional phase space, which has been experimentally reported for a system of compound double pendula with slender bars and asymmetrically oriented square plates as its elements [9,10]. The analysis of the compound double pendulum aids in the design and understanding of multibody physical systems and thus carries considerable importance.

Fourier series analysis of a simple pendulum is used in the determination of transition from linear to nonlinear oscillations [11]. In previous works, the response plot [12], the phase portrait [13], the Poincaré map [10], the power spectrum [14], the Lyapunov exponent [15] and the bifurcation diagram [6] have been used to detect chaos in dynamical systems. Modern approaches of the first Lyapunov indicator, dynamic Lyapunov indicator and smaller alignment index have been verified with classical approaches to determine the chaotic behavior [16]. The transition from the nonlinear regime to the chaotic regime is experimentally determined and the dependency of the behavior on initial energy as well as the orientation of the system is well documented [17].

The Lyapunov exponent is a logarithmic measure for the mean expansion rate per iteration (i.e. per unit time) of the distance between two infinitesimally close trajectories. Consider a point sequence of the trajectory of the system as $x_0, x_1, x_2, \dots$ and the point sequence of the trajectory for infinitesimally small deviation (δx_0) from the initial condition as $\tilde{x}_0, \tilde{x}_1, \tilde{x}_2, \dots$. Any position of the trajectory can be mapped as a function of its previous state as $x_{n+1} = f(x_n)$. Thus, $\tilde{x}_n = x_n + \delta x_n = f(x_{n-1} + \delta x_{n-1})$. The deviation in the n th step by linear approximation can be written as $\delta x_n = f'(x_{n-1})\delta x_{n-1}$. The logarithmic mean of this deviation is given by

$$\sigma = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{l=0}^{n-1} \ln |f'(x_l)|. \quad (1)$$

If the system is a forced pendulum then the Lyapunov exponent (σ) is zero. If σ is negative then the deviation depletes to zero and is thus insensitive to the infinitesimal changes in the initial conditions. If σ is positive, it suggests the exponential separation of the trajectories for a given initial condition and thus the system exhibits chaotic behavior.

For an n -dimensional dynamical system, a section of $n - 1$ dimensional surface transverse to the trajectory is considered such that for every period, the trajectory of the path traversed by the phase space passes through it. The intersection of the trajectory plotted on this section forms the Poincaré map. This section of trajectory is helpful in identifying the behavior of the system. For an undamped conservative system, a single point on the map implies the periodic behavior, a closed loop pattern implies quasi-periodic behavior, and the irregular distribution of the points indicates chaotic behavior of the system. For a damped autonomous system the points in the phase space move towards an attractor with a recognizable pattern, which signifies the transition of the system from nonlinear motion to linear motion and finally coming to rest. However, irregular distribution of points is displayed when exhibiting chaotic behavior.

Various methodologies are used for tracking the pendulum, which include the usage of an accelerometer [18], a stroboscope [9], an optical encoder [15], a V-scope microcomputer [12], a combination of magnet and coils [19], a sonic rangefinder [20] and video processing by tracking the markers using open-source software or pre-built tools [10]. It is observed that the most preferred methodologies are those which are of non-contact mode for the obvious reason of the possible fluctuation in the experimental results due to the intervention of the sensory equipment.

Dry friction can be defined as the opposing force experienced by a solid object in direct contact with another solid object due to the relative lateral motion, which can be further classified into static friction and kinetic friction depending upon the state of motion. Meanwhile, viscous damping is defined as the opposing force experienced by a solid object due to its interaction with the fluid medium surrounding it. In multibody systems the main contributors to the dry friction are pure dry sliding friction, the stick-slip effect, the Stribeck effect and frictional lag [21,22], the forces due to which are independent of the velocity of the object. The viscous damping forces depend on the surface area of the object as well as its velocity in the medium, and have been found to be proportional to the square of the velocity of the object. Several experimental works have been carried out on the single pendulum accounting for these damping forces where the dry frictional forces are determined to be independent of the velocity and the viscous damping forces are found to be proportional to the square of the velocity [20]. Collectively, these losses are dissipated in the form of heat and sound energy. The Rayleigh dissipation function can be used in the solution to mechanical systems involving the frictional forces depending linearly and nonlinearly on the velocity [23]. Thus, the Rayleigh dissipation function is adopted to account for both the linear damping and the nonlinear damping. The energy dissipated is lumped together by considering the Rayleigh damping constant as a fitting parameter, which is found by comparing the analytical results with the experimental results. The Rayleigh dissipation function has been used as a fitting parameter in modeling various multibody systems, both numerically and experimentally [24].

From the available literature, it can be stated that energy dissipation in a double-pendulum system has not been thoroughly addressed and examined. This has led to the formulation of this work, where a comprehensive experimental study taking into account the energy dissipation is presented. The Rayleigh dissipation function is included in the mathematical formulation and dissipation factors are used as fitting parameters. In this paper, the behavior of the system is experimentally investigated and numerically verified for the entire range of combinations of the primary and the secondary angles in steps of 1° , with an operational range of $0 \leq \theta_0 \leq 180^\circ$ and $0 \leq \phi_0 \leq 180^\circ$ for 32 760 combinations. The transition from the linear regime to the nonlinear regime is decided when the deviation of the simulated results from the results of the linearized equation of motion is within 1% of their trajectory, while the transition from the nonlinear regime to the chaotic regime is determined using phase portraits and Poincaré sections.

A planar compound double pendulum is chosen to determine different behavioral regimes of a chaotic deterministic system because of its rich chaotic behavior, the simplicity in its replication into a physical model, the lower rate of energy loss, the ease of tracking and the solvable mathematical model. The upper and the lower pendula are anodized with white and black color, respectively, for better tracking of the pendula. For the scope of our experiment, a non-contact mode of measurement is imperative and hence the video processing technique is chosen for its minimal and non-additive error in measurement. The pendula are detected as a complete unit based on their individual colors, and their locations are determined with reference to their individual hinge positions. This technique further improves the accuracy of measurement by diminishing the effects of warping of the pendulum in comparison to the cited video processing methods, which use markers as a reference while tracking.

The equations of motion of the system are derived using Lagrangian formulation. As a general closed-form solution is unattainable, numerical solutions are used for analysis. When the initial angles are small, the double pendulum behaves similar to a system of two spring-coupled oscillators and as the initial angular displacement(s) increases, the response of the system gradually transitions to nonlinear behavior and then subsequently to chaotic behavior at increased energy levels. This transition is presented in the right quadrant of the combination of initial angles. A surface equation is fit to the Rayleigh dissipation factors to generalize the relationship between the said dissipation factors and the parameters of the system.

In the further organization of the paper, the governing equations of motion and the scheme of the solution are discussed for the cases of an undamped and damped system of both the single and double pendula. The experimental setup and methodology of measurement of the system of the compound double pendulum used for the experiments are elaborated. Furthermore, the experimental and numerical results are assessed in detail followed by discussions on the analysis of errors in measurement, uncertainty in the classification of regimes, and dominance of viscous damping and dry friction damping at different stages of pendulum oscillation. Finally, a note on the effects of the amplitude difference (ψ) on the duration of oscillation is presented after clearly identifying the amplitudes at which viscous and dry friction effects are predominant.

2. Mathematical model

The mathematical model for the undamped single pendulum is studied by linear stability analysis with a brief explanation of the basic terminologies used in the nonlinear analysis along with a brief discussion on the behavior of the damped simple pendulum. The linear normal modes (LNMs) of the double pendulum are determined and extended to the nonlinear normal modes (NNMs). The equations of motion of the undamped and damped compound double pendulum are derived using the Lagrangian formulation. The energy dissipation in the physical system of the double pendulum is experimentally observed in later sections of the paper.

2.1. Case of simple pendulum

The phase space of the simple pendulum forms the basis for understanding the terminologies used in nonlinear dynamics within the scope of this work. We consider an undamped simple pendulum with a point mass m , which is attached to a fixed point with a string of fixed length l having negligible mass. The angle made by the pendulum with a vertical axis measured in the anticlockwise direction is considered to be θ . The differential equation representing its motion is $\ddot{\theta} + (g/l) \sin \theta = 0$, where g is the acceleration due to gravity. The kinetic energy is

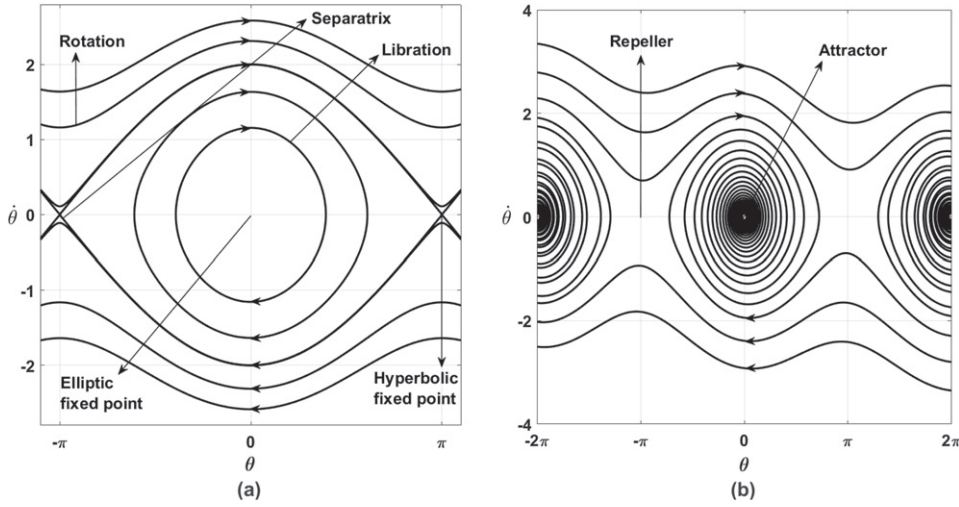


Figure 1. Phase space of an autonomous simple pendulum. Subplot (a) represents the case of an undamped system where the origin forms the stable/elliptic fixed point. Within the periodic orbits around the origin, the system is said to be in libration. The open curves represent the state of rotation. The points at $(\pm\pi, 0)$ form hyperbolic/saddle fixed points, which represent the state of the vertical position. The periodic orbit passing through the saddle points and separating the open curves from the closed curves is called the separatrix. Subplot (b) represents the damped system released from initial angular displacement of θ with an initial zero angular velocity. The origin forms the attractor where the system will eventually come to rest. The vertical position of the pendulum forms the repeller in a damped simple pendulum.

$K = (ml^2\dot{\theta}^2)/2$, and the potential energy is $P = -mgl \cos \theta$. The angular momentum of the system will be $p = ml^2\dot{\theta}$. Thus, the Hamiltonian can be given as $H(\theta, p) = (p^2/2ml^2) - mgl \cos \theta$, which is the total energy of the system written as a function of the two canonically conjugate variables, the angular position θ and the angular momentum p . Thus, the equation of motion can also be written as $\dot{\theta} = (\partial H / \partial p) = (p/ml^2)$ and $\dot{p} = -(\partial H / \partial \theta) = -mgl \sin \theta$. The fixed-point solutions for the system are found to be $(\theta, p) = (n\pi, 0)$, where n is an integer. As the Hamiltonian does not depend explicitly on time, the undamped simple pendulum forms a conservative system and thus constitutes periodic orbits in the phase space, represented by closed elliptic curves in the phase portrait of the system as shown in figure 1(a).

In figure 1(a), the hyperbolic fixed points or saddle points can be observed when $(\theta, p) = ((2n - 1)\pi, 0)$, where n is an integer, which represents the state of vertical position of the pendulum. A saddle point is a fixed point that acts as an attractor for some trajectories and as a repeller for the others. The elliptic fixed points or the stable fixed points can be observed when $(\theta, p) = (2n\pi, 0)$, where n is an integer. When the total energy of the system is less than mgl , ellipses are observed in the phase space around the elliptic fixed point and the system is said to be in libration, where the average momentum of the system is zero. When the total energy of the system is greater than mgl , the open curves are observed in the phase space and the system is said to be in rotation. The separatrix divides the closed curves and the open curves. It is observed when the total energy of the system is mgl , and passes through both the hyperbolic fixed points.

For the system of a damped simple pendulum, the governing differential equation takes the form $\ddot{\theta} + \gamma\dot{\theta} + (g/l) \sin \theta = 0$, where γ is the damping constant of the system. Figure 1(b)

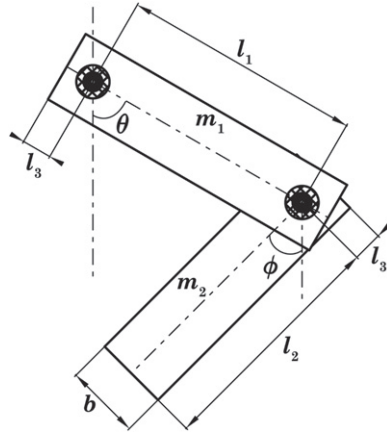


Figure 2. Lumped model of the double pendulum indicating the parameters ($l_1, l_2, l_3, b, m_1, m_2$) and angular positions (θ, ϕ) of the assembly.

represents the phase portrait for a damped simple pendulum, released from an angle of θ with varying initial angular velocity. The system will eventually come to rest at fixed points known as attractors, which are located at $(\theta, p) = (2n\pi, 0)$, where n is an integer. The set of all initial states approaching the attractor over a long period of time ($t \rightarrow \infty$) is called the basin of the attractor. In a nonlinear dissipative system when the attractor is a closed orbit rather than a point then it is called a limit cycle. Contrary to an attractor, a repeller is a fixed point that repels all the trajectories moving towards it and thus no trajectories will pass through it. The set of fixed points with zero angular velocity having angular positions with odd multiples of π constitutes repellers.

2.2. Undamped double pendulum

This discussion is extended to a system of an undamped double pendulum with distributed mass as represented in figure 2. The angular displacements are θ and ϕ for the upper and lower pendula, respectively. The total kinetic energy (E_K) is given by

$$E_K = \frac{1}{24} \left[\dot{\theta}^2 (m_1 \alpha + 12m_2 l_1^2) + \dot{\phi}^2 m_2 \beta + \dot{\theta} \dot{\phi} (12\gamma \cos \psi) \right], \quad (2)$$

where $\alpha = b^2 + 4l_1^2 + 4l_3^2 + 4l_1 l_3$, $\beta = b^2 + 4l_2^2 + 4l_3^2 - 4l_2 l_3$, $\gamma = m_2 l_1 (l_2 - l_3)$ and $\psi = \theta - \phi$. The potential energy (E_P) is given by

$$E_P = \frac{1}{2} \left[(m_2 \delta + m_1 l_1) g - (m_1 + 2m_2) l_1 g \cos \theta - \frac{\gamma}{l_1} g \cos \phi \right], \quad (3)$$

where $\delta = l_2 - l_3 + 2l_1$. The Lagrangian ($\mathcal{L}$) is defined as $\mathcal{L} = E_K - E_P$. The Lagrangian formulation for an undamped double pendulum is as follows:

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{1}{12} \left[\dot{\theta} (m_1 \alpha + 12m_2 l_1^2) + 6\dot{\phi} (\gamma \cos \psi) \right]$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{1}{12} \left[\dot{\phi} m_2 \beta + 6\dot{\theta} (\gamma \cos \psi) \right]$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -\frac{1}{2} [(m_1 + 2m_2)l_1 g \sin \theta]$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = -\frac{1}{2} \left[\frac{\gamma}{l_1} g \sin \phi \right]$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = \frac{1}{12} \left[\ddot{\theta}(m_1 \alpha + 12m_2 l_1^2) + 6\ddot{\phi}(\gamma \cos \psi) - 6\dot{\phi}\dot{\psi}(\gamma \sin \psi) \right]$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) = \frac{1}{12} \left[\ddot{\phi}m_2 \beta + 6\ddot{\theta}(\gamma \cos \psi) - 6\dot{\theta}\dot{\psi}(\gamma \sin \psi) \right].$$

The equations of motion can be derived as

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0, \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0$$

$$\ddot{\theta}(m_1 \alpha + 12m_2 l_1^2) + \ddot{\phi}(\gamma \cos \psi) - \dot{\phi}\dot{\psi}(\gamma \sin \psi) + 6(m_1 + 2m_2)l_1 g \sin \theta = 0$$

$$\ddot{\phi}m_2 \beta + \ddot{\theta}(\gamma \cos \psi) - \dot{\theta}\dot{\psi}(\gamma \sin \psi) + 6\frac{\gamma}{l_1} g \sin \phi = 0.$$

The above equations of motion can be rearranged and written independently with respect to $\ddot{\theta}$ and $\ddot{\phi}$ as follows:

$$\ddot{\theta} = \frac{A_1 + B_1 - C_1}{(12m_2 l_1^2 + m_1 \alpha)m_2 \beta - 36\gamma^2 \cos^2 \psi}, \quad (4)$$

where

$$A_1 = \dot{\theta}^2(-18\gamma^2 \sin 2\psi) - \dot{\phi}^2(6m_2 \beta \gamma \sin \psi)$$

$$B_1 = \left(6\frac{\gamma}{l_1} g \sin \phi \right) 6\gamma \cos \psi$$

$$C_1 = [6(m_1 + 2m_2)l_1 g \sin \theta] m_2 \beta$$

$$\ddot{\phi} = \frac{A_2 + B_2 - C_2}{(12m_2 l_1^2 + m_1 \alpha)m_2 \beta - 36\gamma^2 \cos^2 \psi}, \quad (5)$$

where

$$A_2 = 18\dot{\phi}^2 \gamma^2 \sin 2\psi + 6\dot{\theta}^2 \gamma \sin \psi (12m_2 l_1^2 + m_1 \alpha)$$

$$B_2 = [6(m_1 + 2m_2)l_1 g \sin \theta] 6\gamma \cos \psi$$

$$C_2 = \left(6\frac{\gamma}{l_1} g \sin \phi \right) (12m_2 l_1^2 + m_1 \alpha).$$

The equations of motion (4) and (5) are linearized to determine the LNM. The linearized equations of motion reduces to

$$\ddot{\theta} = \frac{B_1 - C_1}{\underbrace{(12m_2l_1^2 + m_1\alpha)m_2\beta - 36\gamma^2}_{e_1}}, \quad (6)$$

where

$$\begin{aligned} B_1 &= \underbrace{36\gamma^2 \frac{g}{l_1}}_{a_1} \phi \\ C_1 &= \underbrace{(6m_1m_2\beta l_1 g + 12m_2^2\beta l_1 g)}_{c_1} \theta \\ \ddot{\phi} &= \frac{B_2 - C_2}{\underbrace{(12m_2l_1^2 + m_1\alpha)m_2\beta - 36\gamma^2}_{e_1}}, \end{aligned} \quad (7)$$

where

$$\begin{aligned} B_2 &= \underbrace{[36(m_1 + 2m_2)\gamma l_1 g]}_{a_2} \theta \\ C_2 &= \underbrace{\left(72m_2\gamma l_1 g + 6m_1\gamma\alpha \frac{g}{l_1}\right)}_{c_2} \phi. \end{aligned}$$

The linearized equation can be represented in matrix form as

$$\begin{Bmatrix} \ddot{\theta} \\ \ddot{\phi} \end{Bmatrix} = \frac{1}{e_1} \begin{bmatrix} -c_1 & a_1 \\ a_2 & -c_2 \end{bmatrix} \begin{Bmatrix} \theta \\ \phi \end{Bmatrix}. \quad (8)$$

The small-angle approximation of the system of an undamped compound double pendulum is analogous to the system of two coupled oscillators on a smooth horizontal surface. The coupled oscillator constitutes a mass m_1 connected to the left rigid wall by a spring of spring constant k_1 and another mass m_2 connected to the right rigid wall by a spring of spring constant k_3 . The two masses are coupled together by a spring of spring constant k_2 . The linearized equation (8) is analogous to (9), which is the governing differential equation of the coupled oscillators.

$$\begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} = \begin{bmatrix} \frac{k_1 + k_2}{m_1} & \frac{-k_2}{m_1} \\ \frac{-k_2}{m_2} & \frac{k_2 + k_3}{m_2} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}. \quad (9)$$

The state of an oscillating system wherein all its constituents move sinusoidally with the same frequency having fixed phase relation forms the normal mode of that system. Considering the linearized equation of the undamped compound double pendulum (8), the pendula can be

assumed to have the same frequency in order to find its LNM. Thus the angular positions are assumed to be following $\theta(t) = A_1 e^{i\omega t}$ and $\phi(t) = A_2 e^{i\omega t}$. Substituting these values in (8),

$$\frac{1}{e_1} \begin{bmatrix} \omega^2 e_1 - c_1 & a_1 \\ a_2 & \omega^2 e_1 - c_2 \end{bmatrix} \begin{Bmatrix} A_1 \\ A_2 \end{Bmatrix} = 0. \quad (10)$$

To get the solutions for the LNM the determinant of the left-hand side matrix from (10) are found. The undamped natural frequencies of the LNM are found to be $\omega_{n_1} = 1.152$ Hz and $\omega_{n_2} = 2.199$ Hz and the eigenvectors are found to be $[0.552\ 51]^T$ and $[-0.296\ 31]^T$. In the first normal mode both the pendula oscillate with an amplitude ratio of $A_1/A_2 = 0.5525$. In the second normal mode the pendula oscillate out of phase with an amplitude ratio of $A_1/A_2 = -0.2963$. The pendulum is simulated for initial amplitudes proportional to the respective eigenvector values. Figures 3(a) and (b) represent the response plots of the system of a double pendulum in linear oscillations for in-phase and out-of-phase oscillations, respectively, while figures 3(c) and (d) represent the respective configuration space (ϕ vs θ). When the system is in the state of LNM, every component of the system oscillates with the same frequency as observed in figures 3(a) and (b) and has the same displacement ratio as inferred from figures 3(c) and (d), where the slope of the displacements is constant.

At higher energy levels, which are associated with higher amplitudes, the undamped double pendulum exhibits a nonlinear relationship between angular positions and time. As an extension to the LNM, the NNM is defined to be the state of a conservative system where its constituents exhibit synchronous oscillations, i.e. all the constituents of the system reach their respective amplitudes and pass through the zero displacement position simultaneously [25]. Figures 3(e) and (f) represent the nonlinear motion of the system for in-phase and out-of-phase oscillations, respectively, where synchronous oscillations of both the primary (in red) and secondary (in blue) pendula are observed and the corresponding configuration spaces are observed in figures 3(g) and (h). When the system is in the state of NNM, the nonlinear modal lines are observed in the configuration space as observed in figures 3(g) and (h), which is due to the nonlinear relationship of the individual displacements of the pendula with time.

2.3. Double pendulum with dissipation

The energy dissipation due to viscous and dry friction effects is accounted for by the Rayleigh dissipation function ($\tilde{D}$). The Rayleigh dissipation function is given by $\tilde{D} = \frac{1}{2} \sum_{i=1}^2 \xi_i \dot{\theta}_i^2$, where $\xi_1 = \xi_1(\eta, \mu)$ and $\theta_1 = \theta$ are the Rayleigh dissipation factor and angular displacements corresponding to the primary pendulum, respectively, and $\xi_2 = \xi_2(\eta, \mu)$ and $\theta_2 = \phi$ are the Rayleigh dissipation factor and angular displacements corresponding to the secondary pendulum, respectively. ξ_i is the dissipation factor indicative of the combined dissipation due to viscous effects and dry friction. The term η is the viscosity of the surrounding medium and μ is the coefficient of dry friction. The Rayleigh dissipation function has a unit of energy while the dissipation factor has a unit of $\text{kg m}^2 \text{s}^{-1}$. By the inclusion of the non-conservative energy into the Lagrangian formulation with the Lagrangian ($\mathcal{L}$) given as $\mathcal{L} = E_K - E_P$, the Lagrangian formulation takes the form

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} \right) = \frac{\partial \mathcal{L}}{\partial \theta_i} - \frac{\partial \tilde{D}}{\partial \dot{\theta}_i}. \quad (11)$$

The equations of motion are expressed as follows:

$$\ddot{\theta} = \frac{A_1 + B_1 - C_1}{(12m_2 l_1^2 + m_1 \alpha) m_2 \beta - 36\gamma^2 \cos^2 \psi}, \quad (12)$$

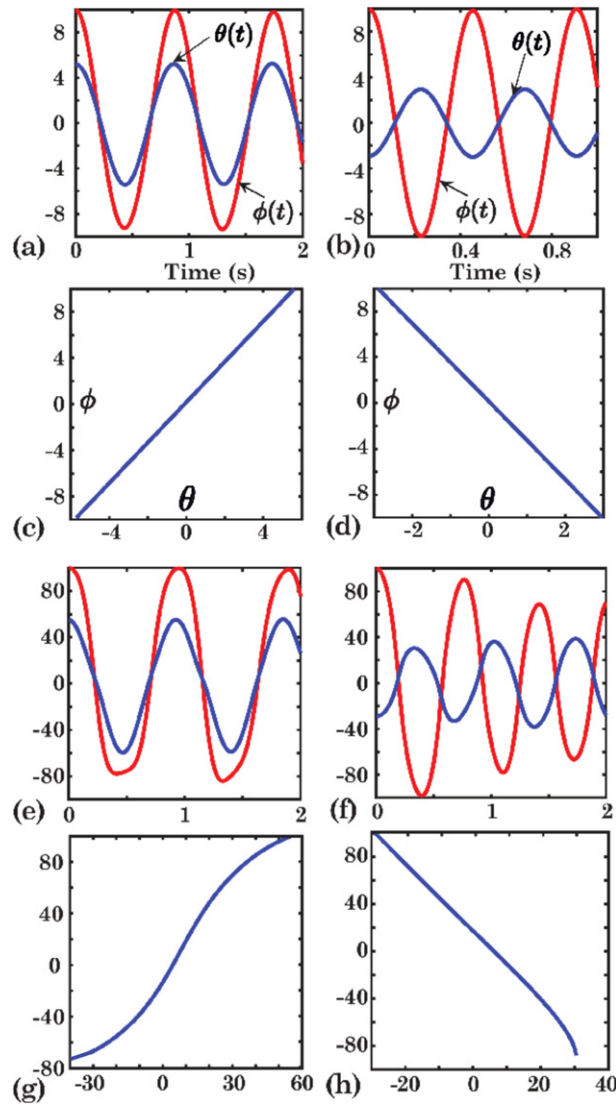


Figure 3. Response plots and configuration space (ϕ vs θ) used to explain the LNM and NNM of an undamped double pendulum. Subplots (a) and (c) represent the response plots and the corresponding in-phase LNM, respectively, for the initial conditions of ($\theta_0 = 5.5^\circ$, $\phi_0 = 10^\circ$). Subplots (b) and (d) represent the response plots and the corresponding out-of-phase LNM, respectively, for the initial conditions of ($\theta_0 = -2.9^\circ$, $\phi_0 = 10^\circ$). Subplots (e) and (g) represent the response plots and the corresponding in-phase NNM, respectively, for the initial conditions of ($\theta_0 = 55^\circ$, $\phi_0 = 100^\circ$). Subplots (f) and (h) represent the response plots and the corresponding out-of-phase NNM, respectively, for the initial conditions of ($\theta_0 = -29^\circ$, $\phi_0 = 100^\circ$). The red- and blue-colored response plots correspond to the angular displacements of the primary and secondary pendula, respectively.

where

$$\begin{aligned}
 A_1 &= \dot{\theta}^2(-18\gamma^2 \sin 2\psi) - \dot{\phi}^2(6m_2\beta\gamma \sin \psi) \\
 B_1 &= \left(6\frac{\gamma}{l_1}g \sin \phi + 12\xi_2\dot{\phi}\right) 6\gamma \cos \psi \\
 C_1 &= \left[6(m_1 + 2m_2)l_1g \sin \theta + 12\xi_1\dot{\theta}\right] m_2\beta \\
 \ddot{\phi} &= \frac{A_2 + B_2 - C_2}{(12m_2l_1^2 + m_1\alpha)m_2\beta - 36\gamma^2 \cos^2 \psi}, \tag{13}
 \end{aligned}$$

where

$$\begin{aligned}
 A_2 &= 18\dot{\phi}^2\gamma^2 \sin 2\psi + 6\dot{\theta}^2\gamma \sin \psi(12m_2l_1^2 + m_1\alpha) \\
 B_2 &= \left[6(m_1 + 2m_2)l_1g \sin \theta + 12\xi_1\dot{\theta}\right] 6\gamma \cos \psi \\
 C_2 &= \left(6\frac{\gamma}{l_1}g \sin \phi + 12\xi_2\dot{\phi}\right) (12m_2l_1^2 + m_1\alpha).
 \end{aligned}$$

These equations are solved numerically using the implicit method(s) for the simulation of the numerical results.

3. Experimental model and methods

The experimental setup shown in figure 4(a) consists of a compound double pendulum made from aluminum, which is firmly held by an inverted L-shaped frame and fixed to a rectangular base. The assembly consists of a primary (upper) pendulum made of two parallel bars, supporting a secondary (lower) pendulum made of a single bar. Two deep-groove ball bearings are press-fitted to the individual bars of the primary pendulum, through which a stud of diameter 8 mm is press-fitted, which is further secured to the L-frame using a fastener. The secondary pendulum is coupled to the primary pendulum by a stud of 8 mm diameter, which runs by the axis of rotation of the bearing of the secondary pendulum and through the lower holes of the primary pendulum. The axis of rotation for both the pendula lies at a distance of 20 mm from their respective free ends.

The deep-groove ball bearings used in the setup are made of stainless steel with an outer diameter of 22 mm, an inner diameter of 8 mm and a thickness of 7 mm. The balls used are of diameter 4 mm, which are rated to withstand a static load of 1.37 kN and a dynamic load of 3.45 kN [26]. The limiting speed of the bearings is rated at 48 000 rpm. For steel and steel material contact, the rolling friction coefficient is rated as 0.001–0.002 and for the clean and dry conditions the static friction coefficient is rated as 0.5–0.8 and the dynamic friction coefficient is rated as 0.42.

The pendula are free to rotate in a two-dimensional plane about the axis along with the studs. Angular measurements are laser engraved in a circular fashion on the surface of the pendulum about its axis of rotation, along with a fixed pointer pointing the instantaneous angle. The operational range(s) of the pendula are $0 \leq \theta_0 \leq 180^\circ$ and $0 \leq \phi_0 \leq 180^\circ$. The distance between the two hinges in the primary pendulum (l_1), the distance from the second hinge to the free end of the secondary pendulum (l_2), the shortest distance from the hinges to the respective free ends of the pendula (l_3), the width of the pendula (b), the thickness of the individual bars

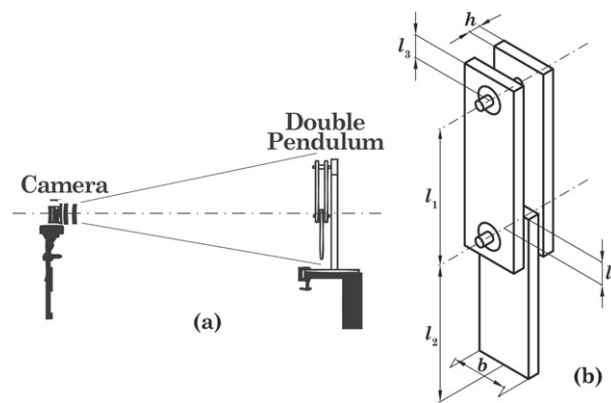


Figure 4. (a) Schematic of the experimental setup. (b) Geometric model of the assembly of the pendula.

Table 1. Geometric dimensions of the components of the double-pendulum setup.

Parameter	Dimensional unit
l_1	160 mm
l_2	140 mm
l_3	20 mm
b	48 mm
h	10 mm
m_1	0.555 kg
m_2	0.215 kg

of the pendula (h), the collective rotary mass of the primary pendulum (m_1) and the collective rotary mass of the secondary pendulum (m_2) are listed in table 1 and represented in figure 4(b).

The rectangular base of the experimental setup is firmly clamped to a rigid slab. The background of the setup is covered with red-colored cloth for better contrast while tracking. The primary pendulum is white anodized while the secondary pendulum is black anodized to aid in tracking them independently. The orientation of the DSLR camera is calibrated using a tripod stand, ensuring that the optical axis passes through the axis of the primary hinge. The focal length of the lens is adjustable in the range 18–55 mm with an aperture range of $f/3.5$ – $f/5.6$. The motion of the pendulum can be captured within the range of 24–60 fps and the shutter speed can range between $1/30$ – $1/4000$ s as per the camera specifications. The positional data are determined by tracking the pendula using image processing, the flowchart of which is illustrated in figure 5.

The input video is in .mp4 format. The upper hinge position is manually determined and the image is smoothened using a bilateral filter so as to retain the edges of the pendula. Two copies of each frame are considered independently to threshold the pixel values representing the upper and the lower pendula. The threshold values are manually determined by iteration until the individual frames account for the majority of the pixels of the respective pendula. This is followed by binarization to convert the region of the respective pendula to white pixels on a black background. Unwanted voids and noises are replaced by using morphological operations. These binary images are used to determine the centroid positions of the individual

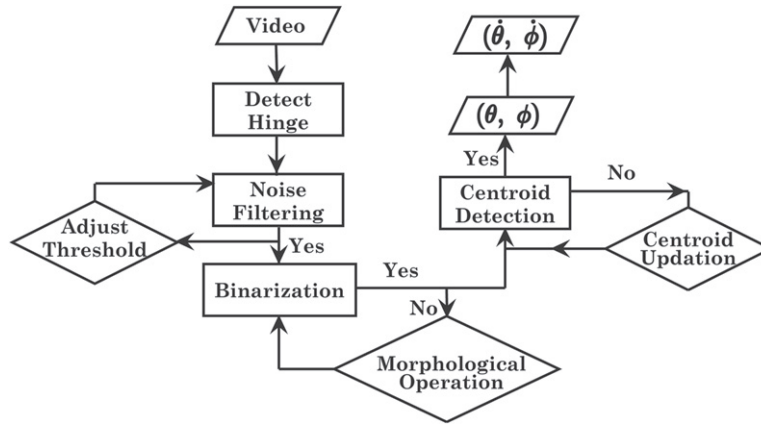


Figure 5. Flowchart of the algorithm for tracking the individual pendula, where the video of the experimental trial is processed using image processing techniques and the parameters $(\theta, \phi, \dot{\theta}, \dot{\phi})$ are determined by detecting the hinge positions and centroids of individual pendula and post-processing these readings.

pendula. When the secondary pendulum is completely eclipsed, the position of its centroid is updated with that of the primary pendulum. The secondary hinge position is determined based on the length of the primary pendulum. The instantaneous angular positions of the individual pendula are determined by calculating the arctan function of the individual centroid positions with the respective hinge positions. The instantaneous angular velocities of the pendula in the current frame are determined by numerical differentiation of the angular positions from the neighboring frames using the formulae $\dot{\theta}_i = (\theta_{i+1} - \theta_{i-1}) / (t_{i+1} - t_{i-1})$ and $\dot{\phi}_i = (\phi_{i+1} - \phi_{i-1}) / (t_{i+1} - t_{i-1})$, where t is the total time elapsed from the instant of releasing the system until the i th frame of the video. The velocity for the boundary cases is considered to be the same as the respective adjacent instantaneous velocities. The complete sequence is scripted using MATLAB software.

4. Results

The experiment is conducted in a closed room with a dry bulb temperature of 30 °C, wet bulb temperature of 23 °C and relative humidity of 55%, which has a considerable effect on the viscosity of the air [27]. The maximum frequency of the system during chaos is measured to be 4.4 Hz, which is greater than the undamped natural frequencies ($\omega_{n1} = 1.152$ Hz $\omega_{n2} = 2.199$ Hz) of the system obtained from (8). The minimum limit of the frame rate is determined to be 22 fps [28], thus videos are captured at 25 fps for better computational efficiency, along with a shutter speed of 1/800 s to ensure minimum blurriness [29], which is determined manually to ensure sufficient exposure to sustain a better quality of the video.

The angular positions are mapped with the time steps and the video is reconstructed with the bounding boxes along with the coordinates of the respective centroids as shown in figure 6. The corresponding velocities of the pendula are obtained by finding the spatial derivative from the available positional data. In chaotic motion, the complete interference of the secondary pendulum by the primary pendulum is overcome by replacing the centroid of the secondary pendulum with the primary pendulum. The deviation in the measurement of the position of the secondary pendulum when it is partially interfered is found to be insignificant. There exist other

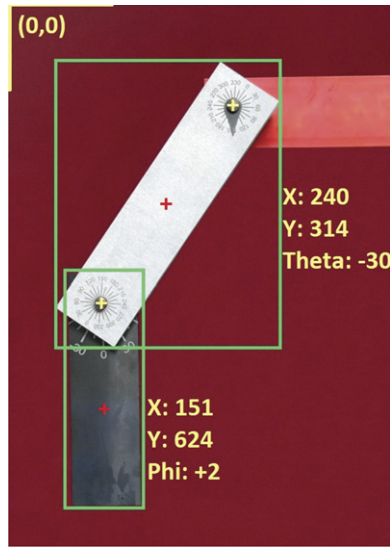


Figure 6. Tracking of the double pendulum with the yellow-colored ‘+’ being the hinge positions and the red-colored ‘+’ being the centroidal positions. The datum for the coordinate measurement is the top left corner of the image.

inevitable sources of error in the video processing technique, which include the errors from improper calibration with the reference plane, rolling shutter effect, insufficient resolution, and unstable illumination. These errors are non-additive in nature, which makes the cumulative error negligible for all purposes within the scope of this work.

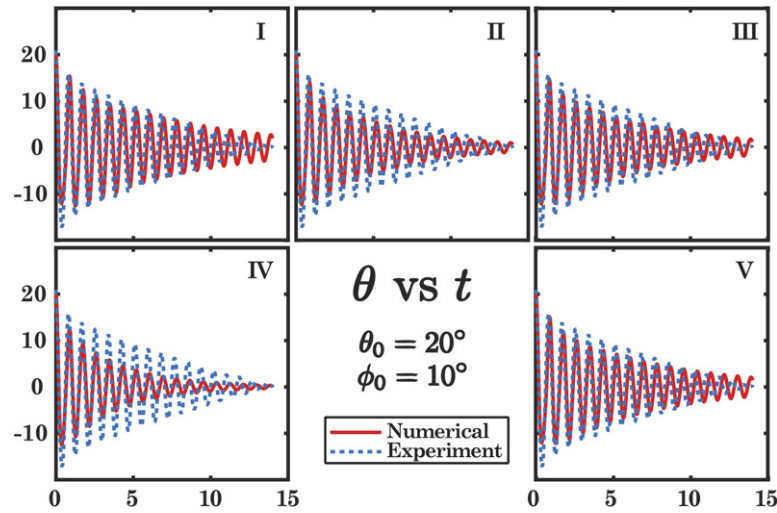
The experimental results of the positional data and the angular velocities of both the pendula are plotted independently with respect to time with a time step of 1/25 fps, i.e. 0.04 s. The optimum values of the dissipation factors are decided by manually tuning the numerical results with respect to the experimental results on the basis of the percentage error in each plot. The average error (ε) is calculated and expressed as a percentage for the angular displacements $\theta(t)$ given by

$$\varepsilon(\theta) = \frac{1}{n} \sum_{t=0}^{t=t_{\text{end}}} \frac{\theta_{\text{exp}} - \theta_{\text{num}}}{\theta_{\text{exp}}} \quad (14)$$

with a time step of 0.04 s, where θ_{exp} and θ_{num} are the experimental and numerical angular displacements of the upper pendulum, respectively, and t_{end} is the time instant at which the pendulum oscillations become zero after complete die-down. Similarly, the average errors $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ corresponding to $\phi(t)$, $\dot{\theta}(t)$ and $\dot{\phi}(t)$, respectively, are determined and presented in table 2. This procedure is illustrated in figure 7 for the combination of initial angular displacements $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$. In case I, the error $\varepsilon(\theta)$ is minimized and is found to be approximately equal to 5% when the dissipation factors are $\xi_1 = 10.507 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and $\xi_2 = 13.493 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$. However, for the same set of dissipation factors the errors $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ are found to be 30%, 9% and 40%, respectively, as shown in table 2. Similarly, in case II, the average error $\varepsilon(\phi)$ is found to be approximately 7% for the dissipation factors $\xi_1 = 17.910 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and $\xi_2 = 21.254 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and the corresponding average errors $\varepsilon(\theta)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ are found to be 11%, 15% and 20%, respectively. For

Table 2. Dissipation factors (in $\text{kg m}^2 \text{s}^{-1}$) for varying cases of minimizing the errors (in %) $\varepsilon(\theta)$, $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$.

Case	Dissipation factor	$\varepsilon(\theta)$	$\varepsilon(\phi)$	$\varepsilon(\dot{\theta})$	$\varepsilon(\dot{\phi})$
I	$\xi_1 = 10.507 \times 10^{-4}$	5	30	9	40
	$\xi_2 = 13.493 \times 10^{-4}$				
II	$\xi_1 = 17.910 \times 10^{-4}$	11	7	15	20
	$\xi_2 = 21.254 \times 10^{-4}$				
III	$\xi_1 = 13.851 \times 10^{-4}$	10	22	4	22
	$\xi_2 = 17.030 \times 10^{-4}$				
IV	$\xi_1 = 26.269 \times 10^{-4}$	15	8	12	10
	$\xi_2 = 30.209 \times 10^{-4}$				
V	$\xi_1 = 15.000 \times 10^{-4}$	6	10	6	18
	$\xi_2 = 15.000 \times 10^{-4}$				

**Figure 7.** Error optimization by calculating percentage error between the numerical results (in red color) and the experimental results (in blue color) by varying the dissipation factors ξ_1 and ξ_2 . The subplots represent the error calculation for θ vs t for the initial conditions of $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$. In case I, the error $\varepsilon(\theta)$ is minimized and is found to be 5%. Similarly in cases II, III and IV, the errors $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ are minimized respectively and the corresponding $\varepsilon(\theta)$ are found to be 30%, 9% and 40%, respectively. Optimum errors are obtained for $\xi_1 = 15.000 \times 10^{-4} \text{ kg m}^2 \text{s}^{-1}$ and $\xi_2 = 15.000 \times 10^{-4} \text{ kg m}^2 \text{s}^{-1}$, for which $\varepsilon(\theta)$ is determined to be 6% and the corresponding θ vs t plot is presented in case V.

case III, the average error $\varepsilon(\dot{\theta})$ is found to be approximately 4% for the dissipation factors $\xi_1 = 13.851 \times 10^{-4} \text{ kg m}^2 \text{s}^{-1}$ and $\xi_2 = 17.030 \times 10^{-4} \text{ kg m}^2 \text{s}^{-1}$ and the corresponding average errors $\varepsilon(\theta)$, $\varepsilon(\phi)$ and $\varepsilon(\dot{\phi})$ are found to be 10%, 22% and 22%, respectively. Finally, for case IV, the average error $\varepsilon(\dot{\phi})$ is found to be approximately 10% for the dissipation factors $\xi_1 = 26.269 \times 10^{-4} \text{ kg m}^2 \text{s}^{-1}$ and $\xi_2 = 30.209 \times 10^{-4} \text{ kg m}^2 \text{s}^{-1}$ and the corresponding average errors $\varepsilon(\theta)$, $\varepsilon(\phi)$ and $\varepsilon(\dot{\theta})$ are found to be 15%, 8% and 12%, respectively.

From figure 7, the conclusion can be drawn that the minimization of $\varepsilon(\theta)$ results in large errors of $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$, as presented in table 2. A similar trend is observed in cases II, III and IV, where errors with respect to ϕ , $\dot{\theta}$ and $\dot{\phi}$, respectively, are at a minimum. An optimum choice for the dissipation factors are $\xi_1 = 15.000 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and $\xi_2 = 15.000 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$, which are determined by tuning the numerical results and presented as case V. Although the errors $\varepsilon(\theta)$, $\varepsilon(\phi)$, $\varepsilon(\dot{\theta})$ and $\varepsilon(\dot{\phi})$ are not as low as their individual minimum values in the first four cases, the cumulative error is relatively less, which is found to be 6%, 10%, 6% and 18%, respectively. However, this procedure is not entirely for chaotic motion as only one set of dissipation factors is chosen to predict the behavior. From table 2 it can be observed that when the error is optimized with respect to $\varepsilon(\theta)$ or $\varepsilon(\dot{\theta})$, the dissipation factors are relatively low when compared to that of the instances of error optimization with respect to $\varepsilon(\phi)$ or $\varepsilon(\dot{\phi})$. Thus, an optimum balance of the dissipation factors is established, albeit by trial and error method.

The results shown in figure 7 present the procedure to fit the dissipation factors for the initial angular displacements $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$. However, at large angular displacements, it is observed that the dissipation factors that minimize the error during the initial 40% of the total oscillation resulted in a large error during the small-angle oscillations. Keeping this in mind, the choice of dissipation factors for large-angle oscillations is based on the experimental findings during the first 40% of the total response. It is observed that for all instances of chaotic motion, the system is found to display chaotic motion during the first 10 s of its release.

Figure 8 shows the numerical and experimental results during the first 10 s of motion for varying initial conditions. Figures 8(a) and (b) are for the case of $\theta_0 = 20^\circ$ and $\phi_0 = 10^\circ$, for which the dissipation factors $\xi_1 = 15.0 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and $\xi_2 = 15.0 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$. Figures 8(c) and (d) show the response curves for the initial conditions of $\theta_0 = 45^\circ$ and $\phi_0 = 140^\circ$. The dissipation factors are found to be $\xi_1 = 6.0 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and $\xi_2 = 6.0 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$. Similarly, figures 8(e) and (f) give the responses of the upper and lower pendula for $\theta_0 = 90^\circ$ and $\phi_0 = 90^\circ$ for which the dissipation factors are $\xi_1 = 4.0 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and $\xi_2 = 6.0 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$. The angular displacements for the initial conditions of $\theta_0 = 45^\circ$, $\phi_0 = 140^\circ$ and $\theta_0 = 90^\circ$, $\phi_0 = 90^\circ$ are obtained numerically without neglecting the nonlinear terms in (12) and (13).

Figures 8(g) and (h) represent the case of $\theta_0 = 120^\circ$ and $\phi_0 = 0$ with the dissipation factors $\xi_1 = 3.4 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and $\xi_2 = 5.4 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$. Figures 8(i) and (j) represent the case of $\theta_0 = 160^\circ$ and $\phi_0 = 160^\circ$ with the dissipation factors $\xi_1 = 5.4 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and $\xi_2 = 4.6 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$. The non-periodic and unpredictable nature of angular displacement ϕ for the initial conditions $\theta_0 = 120^\circ$, $\phi_0 = 0$ and $\theta_0 = 160^\circ$, $\phi_0 = 160^\circ$ shows that the lower pendulum exhibits chaotic motion, which is further verified using phase portraits and Poincaré maps. This procedure is carried out for the entire range of experimental trials; the suitable dissipation factors are determined and the respective behavioral regimes are found. It can be seen from figures 8(g)–(j) that even for chaotic motion, a reasonably good tuning of the dissipation factors is possible. Thus the Rayleigh dissipation function can be used as a convenient fitting parameter to assess damping even in a system with strong nonlinearity.

Phase portraits are presented in figure 9 for the results shown in figure 8, where the blue dotted lines show the experimental results and the solid red line represents the corresponding numerical results, which demonstrate the effective replication of the experimental results and can be further used to distinguish chaotic motion from nonlinear motion. In figure 9(c) a vertical pinch as θ approaches zero for both positive and negative angular velocity $\dot{\theta}$ can be observed experimentally. Similar behavior can be observed numerically here for $\xi_1 = 6.0 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ and $\xi_2 = 6.0 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$ with a reasonably good agreement between the two results. Figure 9(d) shows the phase portrait for the secondary angle

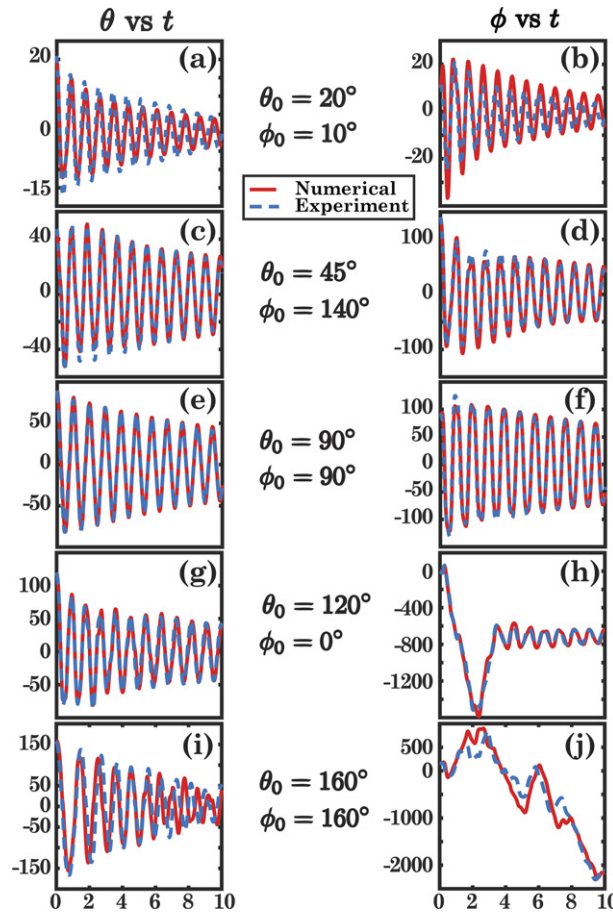


Figure 8. Plots of angular displacement versus time for different initial angle combinations, where the numerical results (in red color) are tuned with the experimental results (in blue color) by suitably finding the dissipation factors using error optimization. The subplots (a) and (b) represent the angular displacement plots of the primary and the secondary pendula, respectively, for the initial angles of $\theta_0 = 20^\circ$ and $\phi_0 = 10^\circ$. Similarly, the subplots (c) and (d), (e) and (f), (g) and (h) and (i) and (j) represent the angular displacement plots for the initial angular displacements of $(\theta_0 = 45^\circ, \phi_0 = 140^\circ)$, $(\theta_0 = 90^\circ, \phi_0 = 90^\circ)$, $(\theta_0 = 120^\circ, \phi_0 = 0^\circ)$ and $(\theta_0 = 160^\circ, \phi_0 = 160^\circ)$, respectively.

ϕ , in which a sudden veering of the curve occurs at the right and left extreme ends for a few initial oscillations; this is also captured in the numerical results for the same combination of dissipation factors.

Figures 9(e) and (f) show the phase portrait for the case of $\theta_0 = 90^\circ$ and $\phi_0 = 90^\circ$. In this case, there is an upward vertical pinch in the trajectory of the phase portrait for the initial few cycles. This is observed when $\dot{\theta}$ is negative. However, in figure 9(f) there is no sudden veering of the trajectory as observed previously in figure 9(d). One possible explanation could be that the nonlinearity in the system is a cumulative effect of $\sin \theta$, $\sin \phi$, $\dot{\theta}^2$, $\dot{\phi}^2$, $\sin(\theta - \phi)$ and $\sin 2(\theta - \phi)$ terms as seen in (12) and (13). Hence, for the initial conditions of $(\theta_0 = 45^\circ, \phi_0 = 140^\circ)$ and $(\theta_0 = 90^\circ, \phi_0 = 90^\circ)$ nonlinearity does exist, albeit in a dissimilar fashion primarily due to the contribution of ψ and 2ψ , where $\psi = \theta - \phi$.

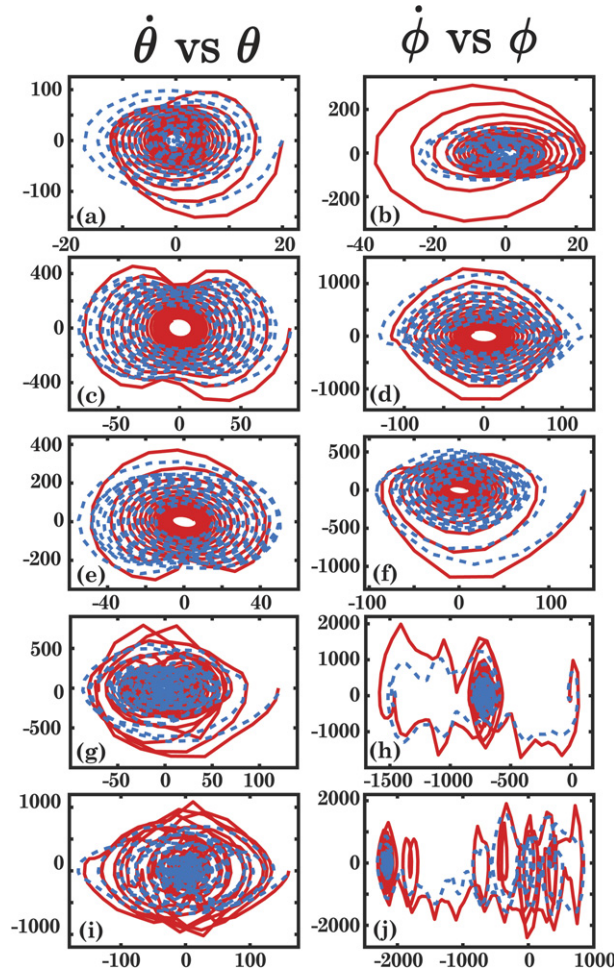


Figure 9. Phase portraits (angular velocity versus angular displacement) for the primary and the secondary pendula for the following set of initial angles, (a) and (b) $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$; (c) and (d) $\theta_0 = 90^\circ$, $\phi_0 = 90^\circ$; (e) and (f) $\theta_0 = 45^\circ$, $\phi_0 = 140^\circ$; (g) and (h) $\theta_0 = 120^\circ$, $\phi_0 = 0$; (i) and (j): $\theta_0 = 160^\circ$, $\phi_0 = 160^\circ$. The red and blue trend represents the numerical and the experimental results, respectively.

Figures 9(g) and (h) present the phase portraits for the case of $\theta_0 = 120^\circ$ and $\phi_0 = 0$. In figure 9(h), multiple basins of attraction are found along $\phi = 0, -1080^\circ$ and -1440° before ultimately settling into the attractor situated along $\phi = -720^\circ$. Figures 9(i) and (j) represent the case of $\theta_0 = 160^\circ$ and $\phi_0 = 160^\circ$. Multiple basins of attraction are observed in figure 9(j) along $\phi = 720^\circ, 360^\circ, 0, -360^\circ, -720^\circ$ and -1800° . Finally, the lower pendulum settles into the attractor at $\phi = -2160^\circ$; this is observed experimentally and supported by the simulation results.

Attractors are found in the phase space for simulated results for every $(\phi, \dot{\phi}) = (2n\pi, 0)$, where n is an integer, for a wide range of dissipation factors. However, for minute variation of dissipation factor values from the tuned values, the trajectory is found to be attracted into multiple basins of attraction. This is demonstrated in the expanded view of the phase portrait

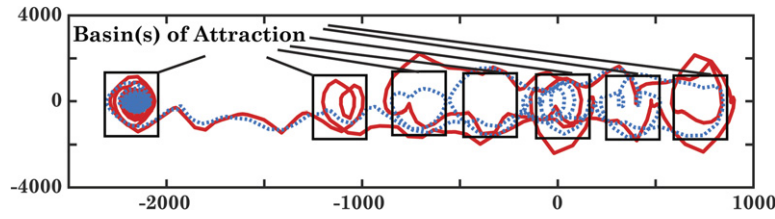


Figure 10. Expanded view of figure 9(j) showing the phase portrait for the secondary pendulum for the initial angles of $\theta_0 = 160^\circ$, $\phi_0 = 160^\circ$. The simulated results (in red) are found to be sensitive to the dissipation factors and thus the trajectories are found to be damping into various basins of attraction for small variation in dissipation factors as shown by the rectangular boxes and thus acting as a basin of attraction.

of figure 9 in figure 10 where the basins of attraction are indicated with rectangular boxes. This phenomenon is also observed with multiple experimental trials for the same initial angles due to the sensitivity of the system in the chaotic regime, as exact replication of the initial conditions cannot be ensured in multiple experimental trials.

Thus, the angular displacements of $(\theta_0 = 120^\circ, \phi_0 = 0)$ and $(\theta_0 = 160^\circ, \phi_0 = 160^\circ)$ are classified as chaotic motion because of the high sensitivity of the trajectory for minute changes in initial conditions. While fitting the numerical results for chaotic motion, the trajectory is found to be going into multiple attractors in multiple trials with varying dissipation factors; however, by careful tuning of the dissipation factors, the same attractor is identified as that of the experimental result and the trajectory traced is found to be unique for a given set of dissipation factors.

Poincaré sections are used to substantiate the chaotic behavior of a system. As the phase space of the system of the double pendulum is of four dimensions comprising the variables $\theta, \phi, \dot{\theta}$ and $\dot{\phi}$, the section of the phase space of any two variables forms the Poincaré map when any one of the other two variables is constant. Figure 11 represents the Poincaré maps, which are plotted for the angular velocity of the secondary pendulum against its angular position for all the time instances when the primary pendulum is rotating in a clockwise direction and is at an angular position of 0 degrees. Thus, the maximum number of experimental data points is accounted for.

Figure 11(a) shows the case of $\theta_0 = 120^\circ$ and $\phi_0 = 0$ and figure 11(b) presents the case of $\theta_0 = 160^\circ$ and $\phi_0 = 160^\circ$. In figure 11(a) the experimental findings and numerical results, which are tuned by a suitable set of dissipation factors, are found to be collapsing onto the same attractor situated at $\phi = -720^\circ$. This is also observed in figure 9(h). Similarly, in figure 11(b), both the experimental and numerical results are found to be converging to the same attractor situated at $\phi = -2160^\circ$, which is again observed in figure 9(j). For a damped oscillation with either periodic or quasi-periodic oscillation, Poincaré maps are found to have a pattern in their appearance when the trajectory moves towards the attractor as seen in the plots shown in figures 11(a) and (b). However, when the system is exhibiting chaotic behavior, map points are dispersed randomly all around the phase plane. For the results shown in figures 11(a) and (b), the map points are spread over a wide range of ϕ , which is of the order of $\sim 10\phi_0$. This is used to characterize the chaotic behavior of the system for angular displacements of $\theta_0 = 120^\circ, \phi_0 = 0$ and $\theta_0 = 160^\circ, \phi_0 = 160^\circ$. Similarly, Poincaré maps are plotted to verify the chaotic behavior for all the experimental results with varying initial conditions to determine the behavioral regimes—linear, nonlinear and chaotic regimes.

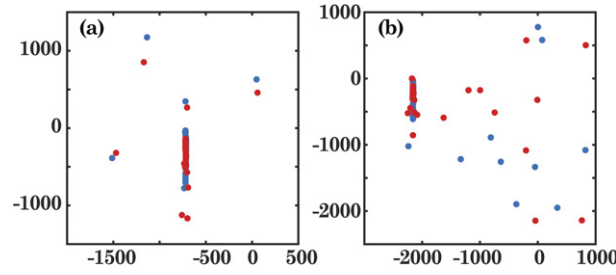


Figure 11. Poincaré maps are plotted for the angular velocity of the secondary pendulum against its angular position ($\dot{\phi}$ vs ϕ) for all the time instances when the primary pendulum is rotating in a clockwise direction and is at an angular position of 0 degrees ($\dot{\theta} < 0$ and $\theta = 0$). The subplot (a) represents the case of initial angles of $\theta_0 = 120^\circ$, $\phi_0 = 0$, and the subplot (b) represents the case of initial angles of $\theta_0 = 160^\circ$, $\phi_0 = 160^\circ$. The red and blue dots represent the numerical and the experimental results, respectively. The region with irregular distribution of points represents chaotic behavior while the region with the points having a pattern which finally settle into an attractor represents the damping of the system from the nonlinear regime to the linear regime and finally coming to rest. In subplot (a) the attractor is found along $\phi = 720^\circ$ and in subplot (b) the attractor is found along $\phi = 2160^\circ$.

When the average deviation of the simulated results from the results of linearized equations of motion (6) and (7) is less than 1%, the instance of the initial conditions results in linear motion. Linear behavior is not in any strict sense because of the existence of damping; however, for the sake of brevity in this article, linearity reflects the linear dependency of the system behavior on θ , ϕ , $\dot{\theta}$ and $\dot{\phi}$. The nonlinearity of the motion is determined for the instances that exhibit average deviation greater than 1% for the results of the nonlinear equation from the linearized equation. The instances with multiple attractor basins sensitive to the initial conditions are classified as chaotic behavior by plotting the phase portraits and hence by extension Poincaré maps.

The outcome of the characterization of linear motion, nonlinear motion and chaotic motion is presented as a behavioral regime classification plot depending upon initial angular displacements in figure 12. The solid black-colored curve running vertically separates the nonlinear regime on the left from the chaotic regime on the right. The region in between the adjacent red-colored lines represents the region of uncertainty where, by experiment, the motion is found to exhibit both chaotic and nonlinear behavior for the same set of initial angular displacements in multiple trials, which is discussed later. The enclosed region at the origin of the graph represents the linear regime. From the regime, it can be inferred that chaotic behavior does exist at a relatively lower energy level (e.g. $\theta_0 = 90^\circ$, $\phi_0 = 0$) as compared to the nonlinear behavior that can be observed at a higher energy level (e.g. $\theta_0 = 90^\circ$, $\phi_0 = 100^\circ$). This is observed due to the existence of optimal combinations of the primary and the secondary angles, where the lack of energy transfer between the primary and the secondary pendulum helps in sustaining nonlinear behavior even at higher energy levels.

In the region of uncertainty as explained in figure 12, for the same initial conditions, the system is found to display both nonlinear as well as chaotic motion in different trials. This behavior is augmented by the phase portraits shown in figure 13. The three cases shown in figure 13 are for initial conditions of $\theta_0 = 113^\circ \pm 0.1^\circ$, $\phi_0 = 117^\circ \pm 0.1^\circ$. The subplots in figure 13(a) having initial angular displacement of ($\theta_0 = 112.9^\circ$, $\phi_0 = 116.9^\circ$) and figure 13(c) having initial angular displacement of ($\theta_0 = 113.1^\circ$, $\phi_0 = 117.1^\circ$) show chaotic behavior with

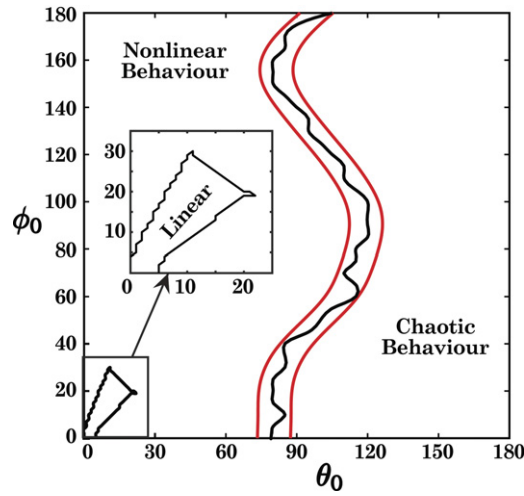


Figure 12. The behavioral regime of the damped autonomous double-pendulum factor-ing range $0 \leq \theta_0 \leq 180^\circ$ and $0 \leq \phi_0 \leq 180^\circ$, in which the set of all initial angles leading to linear motion, nonlinear motion and chaotic motion has been demarcated. The solid vertical black-colored curve separates the nonlinear regime on the left (excluding the enclosed linear regime) from the chaotic regime on the right. The region of uncertainty is bounded by adjacent red-colored lines, where the system is found to exhibit both nonlinear and chaotic behavior experimentally. The initial angular positions displaying linear motion are represented by an enclosed region at the origin.

different basins of attraction for a minute change in initial angles from that of the nonlinear behavior exhibited for the initial conditions of $(\theta_0 = 113^\circ, \phi_0 = 117^\circ)$, whose phase portrait is shown in figure 13(b). Thus, at the boundary of the regime separating the chaotic regime from the nonlinear regime a region of uncertainty is observed in which both nonlinear and chaotic behavior can be observed irrespective of the initial energy and phase difference. This can be attributed to the fact that identical initial conditions could not be replicated over ~ 50 trials. Therefore, the regime boundary is rather difficult to determine with a high degree of accuracy as compared to the accurate regime boundary between the linear and nonlinear regimes.

After analyzing the results from figures 8 and 9, it can be qualitatively concluded that with increasing initial angular displacements, there is a corresponding decrease in the values of the dissipation factors. Hence, as a first-order approximation, the dissipation factor can be written as $\xi_i = \mathcal{F}(\theta_0, \phi_0, \eta, \mu)$, where η is the viscosity of the surrounding medium and μ is the coefficient of dry friction (rolling or sliding or both). Additionally, since there is damping due to viscous drag, the geometric dimensions also play a significant role in the energy dissipation. Since dry friction is also involved at the bearing surface, the masses of the pendula contribute to the energy dissipation as well. The dependency of the dissipation factors ξ_1 and ξ_2 on the initial angular displacements θ_0 and ϕ_0 is depicted in figure 14.

Figures 14(a) and (b) represent the color-mapped surface fit for the dissipation factors ξ_1 and ξ_2 , respectively. The curve-fitting toolbox from MATLAB is used to plot the surface by choosing the ‘thinplateinterp’ option as fitype. Thick red-colored spheres represent the dissipation factors considered to fit the surface. The dissipation factors presented here are chosen by considering the points having sufficient deviation from the neighboring ones, such that the undulations in the surface are clearly captured. From the above plots, it can be stated that the dissipation factors ξ_1 and ξ_2 are relatively low at large and very large initial angular

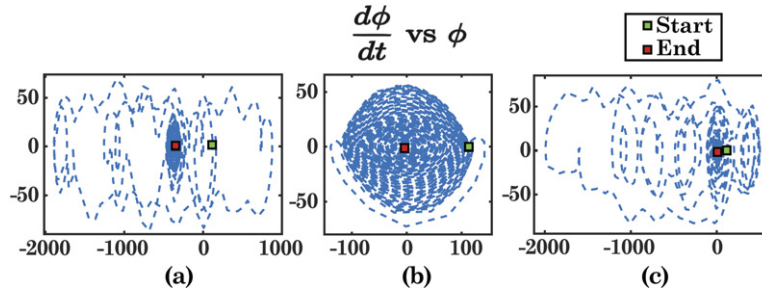


Figure 13. Qualitative uncertainty analysis of the transition between chaotic and non-linear behavior for the initial conditions of $\theta_0 = 113^\circ$, $\phi_0 = 117^\circ$. Subplot (a) represents the phase portrait of the secondary pendulum with initial angular displacements of $\theta_0 = 112.9^\circ$, $\phi_0 = 116.9^\circ$. Similarly, subplots (b) and (c) represent the phase portraits for these secondary pendulum with initial angular displacements of $\theta_0 = 113^\circ$, $\phi_0 = 117^\circ$ and $\theta_0 = 113.1^\circ$, $\phi_0 = 117.1^\circ$, respectively. Subplot (a) displays chaotic behavior even with lower energy and the same initial phase difference, asserting the uncertainty in the boundary separating the nonlinear regime from the chaotic regime.

displacements. However, within the limit of small-angle approximation for $\theta \leq 15^\circ$, there is a significant increase in the values of ξ_1 and ξ_2 . This dependency of dissipation factors on the initial angular displacements, and hence by extension θ and ϕ , was suggested in the previous section, although in a qualitative capacity. In order to quantify the relationship between dissipation factors and initial angular displacements, the surface equation (15) is fit to the results shown in figures 14(a) and (b).

$$Z_i = \bar{A}_i \cos(\bar{B}_i \theta_0) \cos(\bar{C}_i \phi_0) \exp \left(-\sqrt{\frac{\theta_0^2}{\bar{E}_i} + \frac{\phi_0^2}{\bar{F}_i}} \right) + \bar{D}_i. \quad (15)$$

In figures 14(c) and (d) the surface in yellow color represents the surface fit using equation (15). A total of six arbitrary constants are obtained from equation (15). These constants are determined by minimizing the root mean squared errors for ξ_1 and ξ_2 with respect to Z_1 and Z_2 , given by

$$\mathcal{R}(\xi_i) = \sqrt{\frac{\sum_{i=1}^n (Z_i(\theta_0, \phi_0) - \xi_i(\theta_0, \phi_0))^2}{n}}, \quad (16)$$

where $\mathcal{R}$ is the root mean squared error and n is the number of data points considered. The minimized root mean squared errors are $\mathcal{R}(\xi_1) = 1.06 \times 10^{-4}$ and $\mathcal{R}(\xi_2) = 0.84 \times 10^{-4}$. The corresponding arbitrary constants in the surface equation(s) are tabulated in table 3. From figures 14(c) and (d), it can be inferred that both the surface fits show a similar trend in the chaotic regime as well as the nonlinear regime. While there is a significant increase in the values of ξ_1 and ξ_2 near the linear regime, this is primarily due to the significant contribution of dry friction damping. Dry frictional effects are comparatively higher for the upper pendulum because of the higher combined weight acting on its bearing.

The rate of energy dissipation is observed to be higher when viscous effects dominate. This is supported by the nonlinear decay envelope at higher energy bands due to the large angular displacements and angular velocities as seen in figure 15. However, once a considerable amount of energy has been dissipated, the response curves for both θ and ϕ collapse into the linear decay

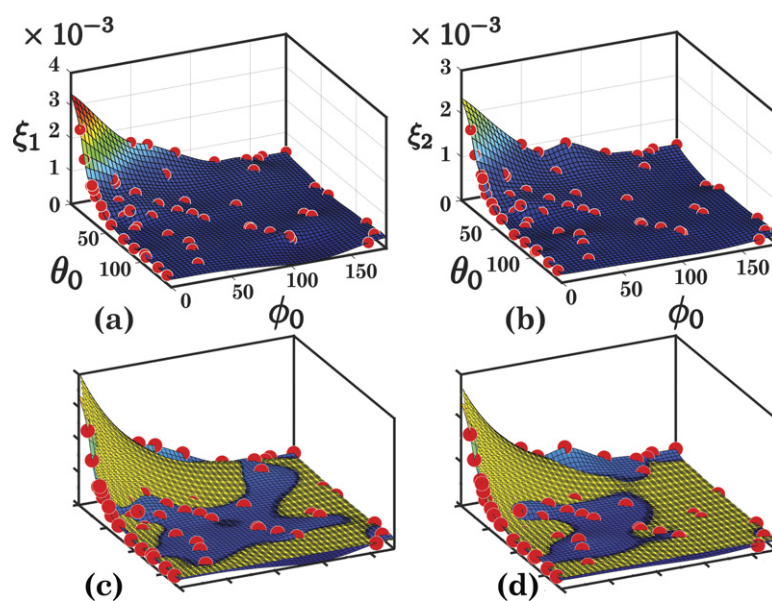


Figure 14. Subplots (a) and (b) represent the surface representing the variation of Rayleigh dissipation factors ξ_1 and ξ_2 , respectively, for different initial angular displacements. The surface color varies from blue to red with the magnitude of the dissipation factors. To quantify the relation between ξ_1 , ξ_2 and the initial angles, surface equation (7) is fit to the ξ_1 and ξ_2 values by minimizing the mean squared error, which is represented as a yellow-colored surface in subplots (c) and (d).

Table 3. Arbitrary constants for the best fit surface for the variation of dissipation factors as a function of initial conditions as presented in (15).

Constants	$i = 1$	$i = 2$
$\bar{A}_i$	0.00370	0.00243
$\bar{B}_i$	1.42	1.40
$\bar{C}_i$	0.01	0.04
$\bar{D}_i$	0.00029	0.00025
$\bar{E}_i$	0.126	0.165
$\bar{F}_i$	0.635	0.635

envelope. At angular displacements less than approximately 12° – 15° , dry friction damping is the significant dissipation mechanism, which is observed to be the oscillations with a linear envelope in their response plots, and which can be observed to the right of the dotted lines in the subplots of figure 15. However, at angular displacements greater than 15° , although dry friction damping exists, the effect of viscosity is significantly higher, as can be seen with nonlinear envelopes to the left of the dotted lines in the subplots of the said figure 15, as viscous damping forces are proportional to the square of the velocity of the pendula. One possible explanation for dry friction damping being significant at lower amplitudes is the inability of the system due to lower velocities to transition from static to kinetic friction. This interplay of viscous and dry friction effects on the energy dissipation in the pendulum and in pendulum systems is an ongoing work.

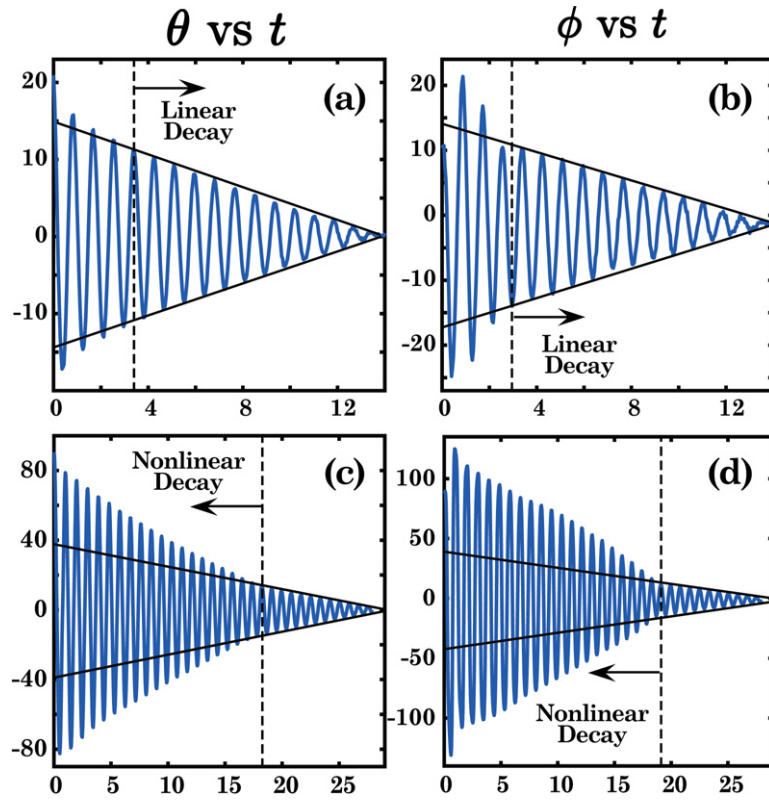


Figure 15. Experimental result of angular displacement versus time plots, where non-linear decay is observed until the amplitudes of 12° – 15° followed by linear decay until complete halt of oscillations, which is distinguished by the vertical dotted line. The damping in the region of nonlinear decay signifies the dominance of viscous damping, and the damping in the region of linear decay signifies the dominance of dry friction damping. Subplots (a) and (b) are representative of initial angular displacements of $\theta_0 = 20^\circ$, $\phi_0 = 10^\circ$ for the primary and secondary pendula, respectively. Subplots (c) and (d) are representative of initial angular displacements of $\theta_0 = 90^\circ$, $\phi_0 = 100^\circ$ for the primary and secondary pendula, respectively.

The total time required for the oscillations to completely die down is found to reduce with the increase in the amplitude difference ($\psi = \theta - \phi$) between the upper and lower pendula. As ψ increases, the secondary pendulum has to rotate independently more often about its axis, which leads to greater energy dissipation and thus, the early die-down in oscillations. A similar observation has been reported by Roy *et al* [30].

Experimental studies have been reported for single pendula, where the damping is nonlinearly dependent on the velocity when the viscous effects are dominant and linearly dependent on the velocity when dry frictional effects are dominant [31]. These observations have been experimentally established in this work for a double-pendulum system under combined viscous and dry friction damping effects. Thus, it can be stated that dissipation is also dependent on angular velocities as well, and can be written as

$$\xi_i = \mathcal{F}(\theta_i, \dot{\theta}_i, \eta, \mu, l_i, b_i, w_i, \rho_i) \quad \text{for } i = 1, \dots, N. \quad (17)$$

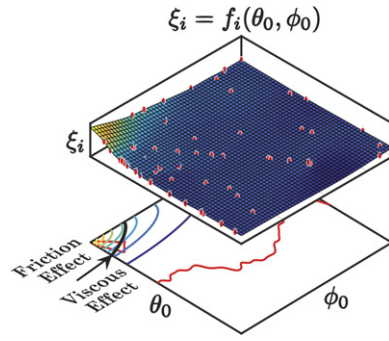


Figure 16. Variation of Rayleigh dissipation factors over the plot of behavioral regime, where the contours of variation of Rayleigh dissipation factors are projected onto the plot of the behavioral regime. The combinations of initial angles where the dry frictional damping is dominant are found to be overlapping the region of the linear regime, while the combinations of initial angles having the dominance of viscous damping are found to be overlapping the nonlinear regime of the behavioral regime plot.

In the above equation θ_i and $\dot{\theta}_i$ are generalized angular displacements and angular velocities, respectively. l_i, b_i, w_i and ρ_i are the length, breadth, width and density of the individual pendula. The understanding of the Rayleigh dissipation function and dissipation factor at present is minimal; however, an initial effort to understand its dependency on angular displacements, angular velocities, viscosity effects, dry friction effects and their influence on behavioral regimes is carried out in this work and presented in figure 16.

The onset of linear decay in energy dissipation as seen in figures 15(a)–(d), which is representative of the damping due to dry friction, corresponds to a sudden increase in the values of Rayleigh dissipation factors shown in figure 16. The contours that are shown in figure 16 are spaced at an interval of $5 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$. The dissipation factor when dry friction effects are dominant is typically in the range of $1.5\text{--}2 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$. Since the Rayleigh dissipation function is a lumped parameter in the context of this work, quantitative analysis of the effects of viscous and dry friction damping individually is a work in progress. While the Rayleigh dissipation factor is dependent on angular displacements and angular velocities, the degree and order of its dependence are currently being explored.

5. Conclusions

An experiment setup of a compound double pendulum is fabricated and image-based object tracking is used to measure the angular displacements directly. A lumped parameter mathematical model is derived by including the Rayleigh dissipation function to complement the experimental results. The Rayleigh dissipation function accounts for the energy dissipation in the double-pendulum system subjected to both viscous and dry friction damping.

The image processing technique is used to track the pendulum, which is of non-contact mode and helps to acquire accurate positional data. The linear, nonlinear and chaotic regime of motion are determined experimentally. The experimental data have been analyzed for the entire range of combinations of primary and secondary angles in increments of 1° in the range $0 \leq \theta_0 \leq 180^\circ$ and $0 \leq \phi_0 \leq 180^\circ$. The linear regime of the system is demarcated when the results of the linearized equation(s) are accurate to within 1% of the results obtained from nonlinear equations of motion.

The numerical results are brought to an agreement with the experimental findings by tuning the numerical results by adopting the Rayleigh dissipation factor(s). A surface equation is fit to the Rayleigh dissipation factor(s) to generalize the relationship between the dissipation factors and the initial angular displacements. Determination of the independent contribution(s) of the viscous and dry friction effects is an ongoing work. The transitional region between chaotic motion and nonlinear behavior is presented with observed uncertainty along the separation boundary.

The dependence of the dissipation factor on the angular velocity of the pendulum needs further examination and is a work in progress. Viscous damping and dry friction damping are considered to be the major contributors in the dissipation of energy. The viscous effects are significant at large angular amplitudes as compared to the dry friction effects that are dominant at small-angle approximations. The response(s) of the pendula show a nonlinear decay when viscous damping is prominent, while the decay in the amplitude is linear when dry friction effects are predominant. The onset of linear decay after a period of nonlinear decay further establishes that energy dissipation under combined viscous and dry friction effects can be separately examined based on the varying amplitude scales.

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